BrickPlan: Diagnosing Future Action Cost and Error Recovery in Visual Assembly Planning

Benchmark research release 2.3
Population model and human studies pending

Abstract

Visual assembly planning requires predicting how an action changes later access, workspace occupancy, and unfinished work. Final-shape accuracy alone does not isolate these abilities. We introduce BrickPlan, an executable benchmark with exact future-cost and recovery labels under a finite assembly abstraction. Original LDraw geometry determines qualified motions and connectivity; a limited rack and persistent off-assembly inspection ledger couple decisions over time. The release contains 1,421 interactive episodes and 7,020 diagnostic questions from 97 source groups. Matched sequences test future legality, inventory-matched fixed contexts test geometric dependence, and delayed-repair questions exclude the goal from recovery credit. Static diagnostics supply four fixed views; active diagnostics charge additional acquisitions. Calibration questions and repeated variants receive explicit scoring treatment. The primary geometry cohort contains 36 pairs from 12 sources, with both exact conflict labels required. Structural controls reveal scope and limits: closure greedy completes 4,163 of 4,263 task-runs, 3,642 optimally; an inverse-last-action control solves 59.0% of recovery questions. We release native-resolution observations, exact evaluators, a question workbench, and a preserved historical model trace. These measurements characterize the instrument; population visual-model and human performance remain unmeasured.

1. Introduction

An assembly action can be legal now and expensive later. Removing a piece may consume the last available holder before a deeper part becomes accessible; replacing another may obstruct unfinished inspection work. If an agent detects an error after several further actions, reversing its latest action need not restore an efficient completion. Measuring these effects requires more than recognizing brick types or comparing a final image.

BrickPlan asks whether an agent can predict the *future action cost* of a visible assembly decision and recover after a costly choice has propagated. We distinguish immediate rejection, a legal action that increases exact remaining cost, and failure to return to an optimal completion corridor. Restricting placements to registered original sites makes these distinctions exhaustively labelable. This deliberate restriction trades manipulation breadth for exact diagnostic targets.

A preceding assembly-only bank illustrates the need for explicit controls. An occupied-blocker closure formula reproduced all 1,802,728 stored state–goal costs. Long solutions alone did not establish deep planning. BrickPlan adds finite external storage and inspection work whose completion persists after reattachment. Static relations remain useful, but occupancy alone no longer specifies the mission state or exact cost.

Our contributions are threefold. First, we define an executable visual assembly task with resource and work obligations, interactive episodes, and explicit checkpoint diagnostics. Second, we construct sequence, geometry, appearance, and recovery controls, with separate static and active observations and auditable scoring units. Third, we release the construction, execution, evaluation, and publication pipeline, including strong structural controls. We do not infer a human–model gap, required internal reasoning depth, or broad physical competence from software verification.

2. Evaluation design

Table 1 fixes the reporting framework before empirical collection. Primary episode outcomes are completion and optimal completion over all assignments, plus excess operations among successes. Diagnostic accuracy is reported by family, scoring role, canonical problem, and source. Joint pair scores require both members correct. Invalid outputs and missing assignments remain in their denominators. Privileged structural policies are separate from image-input results.

The study compares matched assignments and identity seeds with declared action and view budgets. For each VLM, static image and structured conditions share the same initial views, executor, and answer syntax; active acquisition is reported separately. A format/cost pilot precedes the



Figure 1. Primary inventory-matched pair `source-v23-10037-matched-0/geometry/future-00` from held-out source 10037 in the initial isometric view. Both contexts use 63 source pieces (47 fixed); the exact part-number/color multiset, registered poses, start, target, rack, work, and plan match. After 2 shared legal operations, the exact labels are context 1: first conflict at operation 3; context 2: all legal. Only subsets of original fixed instances change. These publication views omit ID overlays; answers are evaluator annotations. Pixel distinction is not perceptual answerability.

Table 1. Experiment framework. Population results are unmeasured; model snapshots and provider settings must be fixed before collection.

Condition	Evidence status
Structural policies and exact search	Measured
SmolVLM-256M, image input	Historical case failed
Open / proprietary VLMs	Pending
Humans, images + registered IDs	Pending
Static / active / structured input	Matched planned study
Stepwise / explicit-plan prompting	Matched planned study

full study. Report snapshot, provider, prompt, image preprocessing, context policy, decoding, calls, measured token use, and acquisition count. Human collection requires documented instructions, training, participant counts, and consent before recruitment.

Identity seeds, paired questions, and variants reuse source geometry. Per-source results and coverage precede pooled claims. 63 source groups are held out from development; the primary geometry cohort covers 9 of them. More diagnostic rows do not create more independent assemblies. Errors are stratified into parsing, immediate legality, future cost, diagnosis, and repair.

3. Related work

LTRON's Break and Make evaluates interactive structural understanding through disassembly and reconstruction [5]. InstructioNet studies construction supported by an agent's own explicit visual instructions [6]. These establish interactive brick construction and long sequences; BrickPlan does not claim either as new. Its registered-site action space permits exact counterfactual completion costs and event-level recovery labels.

LEGO-Puzzles includes both elementary visual questions and a Planning set [4]; it should not be reduced to recognition. PhyBlock combines hierarchical assembly and physical-understanding questions [3]. Brick-Composer introduces BC-Bench for diverse brick selection and pose estimation [2]. BrickPlan fixes placement sites and instead measures resource coupling, future legality, and delayed corridor return. It complements broader spatial tasks.

Shortest-path search and admissible heuristics are established tools [1]. Failure of a weak policy does not prove hardness. We therefore include a strong static relaxation, capacity-aware dispatch, bounded and full A*, and shallow-choice bounds. The contribution is an auditable visual diagnostic instrument, not a new planning algorithm.

4. Task and observation contract

4.1. State, geometry, and actions

Let $A \subseteq V$ be occupied registered sites, $G \subseteq V$ the target, $J \subseteq V$ required inspections, and $D \subseteq J$ completed

inspections. The mission state is $x = (A, D)$. Each piece P_i belongs to one original site S_i. Detached pieces occupy universal oversized rack holders, subject to

$$r(x) = |V \setminus A| \leq K. \tag{1}$$

An `audit` completes a pending job only while its piece is detached; completion persists after replacement. The terminal condition is $A = G,\ D = J$, followed by an explicit `finish`.

An `attach` or `detach` must satisfy qualified straight translation, collision, stud-connectivity, registration, and capacity rules. Fixed pieces form a connected clamped assembly. Executable positions preserve original LDraw transforms. Guarded inset meshes and prescribed paths define qualification; this is not force, gravity, clutch, hand-access, or arbitrary rigid-body simulation. Rack transfer is a resource abstraction. Exploded views are reference diagrams, never executable states.

Each physical operation or audit costs one operation; so does a rejected operation attempt. Camera acquisition and diagnosis have separate budgets. Episodes allow 12 additional views and 12 diagnoses. Public operation budgets depend on visible problem size and work, not private optimum cost.

4.2. Matched observations

Five raster panels show the current assembly, target, all-site reference, registered-piece catalog, and rack. Main views are supersampled at 3840×2880 and reduced to native 1920×1440; catalog tiles are 480×360. Text gives P/S registration, rack inventory, and the inspection ledger. Image input omits pose matrices, installed-site flags, legal-action lists, relation graphs, and exact costs. Explicit registration simplifies correspondence without supplying geometric rules.

Static checkpoint diagnostics provide isometric, left, rear, and right views at revision zero, with later acquisition disabled. Active image diagnostics start with isometric only and charge one acquisition per `inspect`. Structured diagnostics use the same four initial views and additionally provide original matrices, colors, part types, path directions and lengths, and occupancy. They omit the private blocker/support graph; structural software controls receive that graph. Interactive episodes retain their common active-view protocol.

Acquired frames persist with their revision. Free 3D orbit, explosion, and answer review are available in unscored practice only. Human and model policies share the controlled observation and submission contract.

4.3. Diagnostic formats

Table 2 separates interactive execution from checkpoint questions. There are 2,595 next-action questions, 432 ordinary sequence questions, 92 geometry questions, 72 appearance questions, and 3,829 delayed-repair questions. Sequence questions expose the complete proposed plan and ask for its first illegal operation, or `null`. Recovery questions expose the pre-error checkpoint, executed history, current observation, and legal candidates. They ask for a shortest repair step, not hidden model reasoning.

5. Construction and validity controls

5.1. Source groups and sampling

The 412 qualified geometry profiles include 98 base assemblies and 314 context profiles, including zero-yield candidates. They belong to 97 source kit groups. Alternate builds and context variants remain within one group: roadster and kart both belong to 31027. There are 445 development episodes from 34 groups and 976 held-out episodes from 63 disjoint groups.

Version 2.3 preserves the frozen 2.2 definitions and admits 81 of 100 new source candidates; all 19 exclusions remain in the census. Candidate identities, intake strata, source-group splits, and sampling are frozen before new geometry qualification and solver discovery. Every newly admitted base has at least eight registered and eight reachable movable sites. Small historical domains remain regression cases, not new-source admissions. An independent source audit subsequently excludes five candidates whose part-only manifests omit direct model geometry and one with a conflicting root kit identifier. Original files and preliminary admissions remain archived; no replacement candidates are added. Every retained new-source instance is bound to its exact original reference chain, identity, color and pose.

New-source geometry uses deterministic source-order collision pruning and a bounded stud-connected component, preserving every retained original instance and pose. Each admitted source supplies 2 goals, 3 job settings, and 2 starts per mission: 972 seeded episodes in total. Solvability is the only oracle-based admission condition; no policy score, cost gap, or search effort selects these episodes. Legacy mechanism-conditioned tasks and separately constructed geometry diagnostics retain explicit labels. "New source" does not mean absent from pretraining.

5.2. Exact costs and difficulty descriptors

For each mission, we enumerate the capacity-constrained physical component and inspection-bit product. Reverse shortest paths give $C^*(x)$, the minimum remaining operations. The 650 tables contain 61,093,730 state entries and 757,651,278 directed-transition entries, with physical states repeated under different goals and jobs. Checks cover runtime transitions, physical inverses, lower bounds, and Bellman equalities.

Let R be occupied blocker closure needed to remove un-

Table 2. Seven public question families. Episode families are mutually exclusive presentation groups. Every entry has an ID, prompt, constraints, answer format, and scoring rule.

Family	Question	Response and measurement
Assembly / reconfiguration	Transform current assembly into target.	Actions; completion and excess cost
Finite-resource scheduling	Free the holders required for later work.	Actions under a capacity bound
Inspection / persistent state	Complete off-assembly work and restore target.	Actions and audits; physical and ledger completion
Next-action judgment	Which legal choice minimizes completion cost?	Index; any optimum accepted
Future-sequence checking	Where is the first illegal operation?	One-based index or `null`
Geometry / appearance	Does context change, or recoloring preserve, legality?	Conflict answer and joint pair score
Delayed error recovery	Which choice returns to the earlier optimal corridor?	Index on a shortest nonterminal repair

Table 3. Fixed operation-length bins: D1 (1–8), D2 (9–16), D3 (17–24), D4 (25+). They are not cognitive-depth levels.

Tier	Episodes	C^* range	Gap range	A* max
D1	314	1–8	0–0	10
D2	701	9–16	0–4	311
D3	341	17–24	0–10	6,383
D4	65	25–42	0–18	14,307

wanted pieces, insert target pieces, and detach unfinished inspection pieces. A static relaxation is

$$h(x) = |A \triangle G| + 2|R \cap A \cap G| + |J \setminus D|. \qquad (2)$$

Each mismatch needs an operation, each occupied target piece in the required closure needs removal and replacement, and each pending job needs an audit. Capacity and connectivity can add operations. The maximum retained gap $C^* - h$ is 18. Registered sites, rack size, occupancy, jobs, gap, and search effort are separate descriptors; length bins alone do not imply planning difficulty.

5.3. Sequence and shallow-choice controls

The 216 legal/corrupted sequence pairs match first and last actions, action multisets, and prefixes before the first conflict. Legal witnesses contain no repeated-state padding. Each conflict bin (2–5, 6–10, 11+) has 72 pairs. A prefix-only control executes k operations and otherwise answers "all legal." It scores 234/432 at $k = 2$ and 288/432 at $k = 5$; zero-step accuracy is 50.0%. This tests a specific shortcut without establishing immunity to other ones.

At 2,595 of 2,595 next-action checkpoints, the depth-one closure minimizer set includes an optimal choice. Uniform tie selection yields 97.4% expected accuracy; uniform selection among all candidates yields 32.6%. This strong result uses privileged structural relations. We disclose tie sizes, optimistic and pessimistic accuracy, and the matched image/structured comparison rather than infer required reasoning depth.

5.4. Geometry and appearance

The primary cohort contains 36 answer-changing pairs, 18 interventions, and 12 source groups. Both contexts match total count and the exact part-number/color multiset, registered instances and original poses, start, target, rack, jobs, and proposed plan. Only subsets of original fixed instances change; fixed connectivity, full-assembly support, meshes, and paths are separately qualified. No unequal-inventory fallback is allowed. All retained primary failures concern stud support; motion-blocker relations match between contexts.

Of 116 qualified matched interventions, 98 yield no retained future contrast under the declared search. Geometry-only context selection precedes answer-change testing. Retaining answer-changing plans is mechanism conditioning, not representative source sampling. The supplement reports every source, intake stratum, candidate stage, zero-yield intervention, and conflict bin. The 10 earlier unmatched pairs are historical regressions and never enter the primary visual score.

Native certificates compare all five panels in four initial views. Pixel distinction (Figure 1) does not establish perceptual sufficiency or successful active acquisition. The 36 original/grayscale pairs preserve geometry, state, plan, and answer; they test appearance invariance, not independent structures.

5.5. Delayed repair and calibration

For a pre-error checkpoint b, let $\mathcal{C}_b^-$ contain all nonterminal states on any optimal completion from b. After a legal positive-excess error, a distinct-state legal history of lag 1, 3, or 5 remains outside this corridor. At checkpoint z, the repair target is

$$r_b(z) = \min_{y \in \mathcal{C}_b^-} \mathrm{dist}(z, y). \qquad (3)$$

The 3,829 questions have repair costs 1–13, independently verified by forward BFS. Any first action on a shortest repair is accepted. Finishing after a costly detour does not itself earn repair credit.

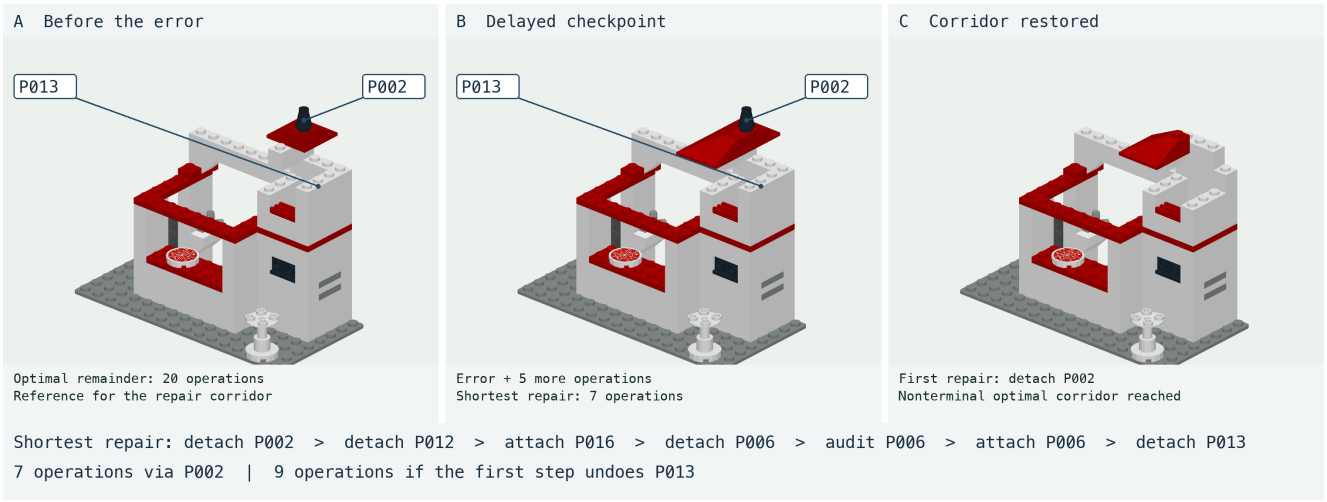


Figure 2. Certified lag-5 repair, `source-10036-35/repair-lag-5`, source 10036. A: the pre-error checkpoint defines the optimal corridor. B: an error and 5 further operations leave it. The last history action is `attach P013`. Undoing it first requires 9 operations to repair; the optimum takes 7, starting with `detach P002`. C: a nonterminal corridor state is restored. These are oracle replay frames, not model reasoning or a model success; exact source instances and actions are archived.

Lag alone does not establish nonlocal repair. Reversing the last public action solves 2,260/3,829 questions (59.0%); the inverse is suboptimal on 1,487 and unavailable on 82. These cohorts are reported separately. All 50 questions with every candidate optimal remain calibration checks and receive no weight in discriminative accuracy.

6. Metrics and scoring units

For a legal unit-cost action with successor x', conditional excess is

$$\delta(x, a) = 1 + C^*(x') - C^*(x) \geq 0. \tag{4}$$

A rejected attempt leaves state unchanged and contributes one excess operation. Thus legal but costly actions remain distinct from immediate invalidity even when rework was already mandatory. Successful regret is $T - C^*(x_0)$ for T charged attempts. Completion and optimal completion use all assignments; mean successful regret accompanies its coverage.

Every legal positive-excess action opens a recovery event. Later errors do not overwrite earlier events. Restoration, immediate and lag-specific cohorts, and right-censored unfinished trajectories are recorded separately. Diagnosis concerns avoidable errors since the last diagnosis; mandatory rework is not an error. Suffix effort is compared with exact remaining cost at diagnosis. Survival curves preserve the first cost-loss event after later recovery, with assigned and unknown counts at each step. The reversible physical components and persistent audits contain no genuine irreversible dead ends; their eligible-choice rate is undefined.

The 7,020 rows represent 6,830 canonical diagnostic problems; repeated statements and appearance variants account for 190 excess rows. Canonical identity includes source geometry, state, target, work, capacity, and the exact question or history. There are 6,970 discriminative assignments. Primary accuracy averages assigned variants within a canonical problem, then problems within a source, then sources equally. Missing variants receive zero before averaging. Row, problem-macro, source-macro, calibration, and recovery-cohort scores remain available. Primary geometry correctness requires *both exact first-conflict labels*, not merely a predicted answer change. Any assigned member makes a pair eligible; a missing partner, absent response, or malformed label earns zero joint credit. Wholly unassigned pairs remain census-only. Repeated rows are averaged within distinct starts, then interventions, then sources. Whole-source bootstrap intervals and leave-one-source-out summaries describe sensitivity conditional on retained sources, not population coverage. The supplement details metric denominators and regret conservation.

7. Measured evidence

7.1. Structural policies

Closure greedy minimizes the static successor bound with visited-state and randomized site tie-breaking. Capacity dispatch prioritizes audits, mandatory removals, safe placements, and temporary staging, with revisited-state escape. Greedy completes 4,163/4,263 runs, with 3,642 optimal; dispatch completes 4,132/4,263, with 3,642 optimal. Completion alone therefore conceals avoidable effort (Table 4).

A* uses graph reopening, $(g+h, h)$ priority, and declared

Table 4. Structural controls on 1,421 episodes. The oracle uses one seed; other policies use three tie-order seeds (4,263 runs). Regret is mean excess among successes. Exact relations are privileged inputs, so these are not image-input VLM results.

Control	Runs	Success	Optimal	Regret	Max exp.
Exact oracle	1421	1421	1421	0.00	—
Closure greedy	4263	4163	3642	1.77	—
Capacity dispatch	4263	4132	3642	1.80	—
Closure A* (100)	4263	4075	4075	0.00	100
Closure A* (1,000)	4263	4199	4199	0.00	1,000
Closure A* (full)	4263	4263	4263	0.00	14,307
Hamming A* (full)	4263	4263	4263	0.00	230,679
Legal random	4263	265	60	13.11	—
Drop inspections	4263	1536	1536	0.00	—
Ignore rack limit	4263	3595	3595	0.00	—

limits without oracle access. Limits of 100 and 1,000 expansions complete 4,075 and 4,199 runs. Full closure A* completes 4,263/4,263, 4,263 optimally, with median 13 and maximum 14,307 expansions. The weaker occupancy-plus-pending-audit heuristic is also reported. These perfect-information measurements constrain difficulty claims; weak controls alone can inflate them.

Dropping inspection obligations or ignoring rack limits produces failures when relaxed plans execute under the actual rules. These are model-mismatch ablations, not strong planners. Raw actions, states, rejections, expansions, timings, and every aggregate are released. Seeds repeat geometry and tie order; cached local timings are not calibrated cross-platform comparisons.

7.2. Reduced-input geometry controls

Count-only, type/color-presence, type/color-count, and sequence-only certificates test explicitly restricted representations. Equal inputs with unequal labels cannot both be correct for a deterministic single-member function of that representation. These label-aware finite-bank ceilings are not fitted held-out predictors, and do not cover instance-sensitive spatial reasoning or policies using pair position as extra information. Separate label-free inventory execution rules and a source-majority predictor fitted only on development labels appear in Table 5. Exact inventories and named occupancy are privileged perception abstractions, not measured visual extraction.

7.3. Historical model integration case

The workbench preserves a version-2.1 SmolVLM-256M-Instruct run on `kart-deck-08` (historical public E069, seed 0). All four calls return scene descriptions instead of JSON actions. Each becomes `INVALID_ACTION`, consumes an operation, and leaves the state unchanged. The controller stops at four calls. This is a failed integration example, not a planning or population result.

The original task, bank hash, image bytes, prompts, raw outputs, parsed actions, feedback, latency, measured MLX token counts, and weight digest are preserved. Native stimuli are 1920 pixels wide; this model's processor resizes the longest edge to 512 with splitting disabled. HD display does not imply native-HD inference. Historical E069 is not relabeled as a version-2.3 experiment. An OpenAI-compatible practice adapter and a separate usage-accounted study runner support further collection.

8. Release and limitations

The React workbench exposes seven families, complete previews, ID/source/ split filters, answer submission, model playback, and review-JSON export. Its isolated WebGL2 renderer preserves original geometry, supports free practice orbit and explosion, and shows part details on hover. Native zoom preserves raster evidence. Dataset cards, public JSONL, sample HD inputs, evaluator dependencies, and deployment files are packaged together. No external hosting URL is claimed.

Software validation replays all 1,421 optimal episodes and 7,020 answers. Audits check sequence multisets, source separation, repair BFS labels, input-pixel differences, budget accounting, missing assignments, overlapping errors, and transcript tampering. Protocol and browser checks cover persistence, observation isolation, and playback. This establishes software behavior, not human usability or real-world mechanical validity.

The principal limits remain structural and empirical. Perfect-information controls characterize this finite bank, not visual planning competence. Registration and inventories simplify grounding. Matched inventories remove the former global-count confound, not every instance-level spatial shortcut. Pixel distinction still needs empirical answerability testing. Geometry excludes forces, arbitrary poses, rotations, simultaneous subassembly motion, and real rack transfers. Source scale does not eliminate dependence, admission bias, or public-CAD contamination. No model

Table 5. Primary inventory-matched geometry: deterministic software rules, not visual performance. Joint correctness requires both exact labels. Intervals resample whole source groups; unavailable estimates are marked unavailable, never inferred as zero.

Software rule	Split	Pairs	Sources	Joint micro	Source macro	95% interval
Named-occupancy inventory	Dev.	12	3	0.0%	0.0%	[0.0, 0.0]
Named-occupancy inventory	Held out	24	9	0.0%	0.0%	[0.0, 0.0]
Public-text inventory	Dev.	12	3	0.0%	0.0%	[0.0, 0.0]
Public-text inventory	Held out	24	9	0.0%	0.0%	[0.0, 0.0]
Development source majority	Held out	24	9	0.0%	0.0%	[0.0, 0.0]

shortsightedness, human–model gap, or immunity to all shortcuts has been measured. The next study must report matched static, active, and structured inputs, source-level uncertainty, and coverage across retained context interventions.

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